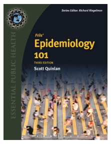

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CHAPTER 3 Epidemiologic Measurements Used to Describe Disease Occurrence
Adjusted Rates

Adjusted Rates

An *adjusted rate* is a rate of morbidity or mortality in a population in which statistical procedures have been applied to permit fair comparisons across populations by removing the effect of differences in the composition of various populations. Rate adjustment can be done for any rate, including incidence rates and mortality rates. This section will review adjustment of mortality rates. A common factor in rate adjustment is age, particularly for mortality rates, which vary greatly by age. Calculation of age-adjusted mortality rates is a much more involved procedure than that required for crude mortality rates. A weighting process entails the use of detailed information about the age structure of the population for which the rates are being age adjusted. There are two techniques for age adjustment of mortality rates. The first is called direct age adjustment. The second is referred to as indirect age adjustment. The direct method uses the age-specific mortality rates for the population in the year of interest and applies them to a standard population with a known age structure. This standard population is usually the United States population in 2000. This method leads to an age-adjusted mortality rate that is the rate expected if the age distribution of the population matched that of our standard population.

The indirect method uses the number of observed deaths in a population over a period of time. Age-specific mortality rates are then identified from a standard population and used to determine how many deaths in the population would have been expected if the mortality rates were the same. The observed and expected deaths are then compared to see how different they are from each other.

The two techniques for age adjustment each have their own advantages and disadvantages. Direct age adjustment allows for a fair comparison of mortality between two populations, but requires that age-specific mortality rates be available for our populations of interest, which may not always be the case. Indirect age adjustment does not have such requirements for age-specific rates, but cannot be used to compare two different populations of interest to each other.

According to the National Center for Health Statistics, the age-adjusted death rate in the United States in 2020 was 835.4 deaths per 100,000 U.S. standard population. **Figure 3.12** compares the age-adjusted mortality rates in the United States by sex and race/ethnicity in 2020 compared to 2019. In 2020, Black males had the highest age-adjusted mortality rates, while White females had the lowest age-adjusted mortality rates.

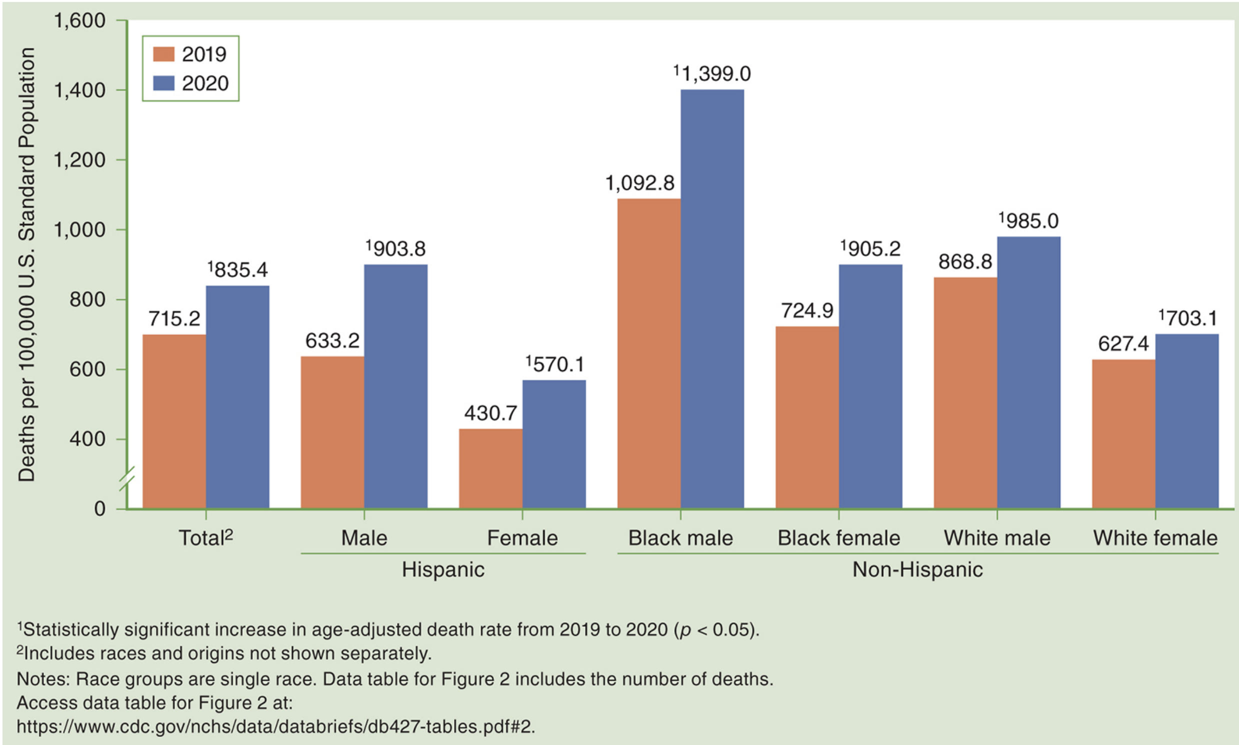

Figure 3.12 Age-adjusted mortality rates, by sex and race and ethnicity: United States, 2019 and 2020

Murphy SL, Kochanek KD, Xu JQ, Arias E. *Mortality in the United States, 2020*. NCHS Data Brief, no 427. Hyattsville, MD: National Center for Health Statistics; 2021.

Description

Returning to the example in which we compared mortality in New York and Utah, assume the crude mortality rate was 914 per 100,000 in New York, and in Utah, the rate was 676 per 100,000. Using direct age adjustment, the corresponding age-adjusted rates were 713 per 100,000 and 816 per 100,000, respectively. The higher crude mortality rate observed in New York in comparison with Utah was due largely to differences in the age structures of the two states, in which New York residents are on average older than Utah residents. You can see that when the rates were age adjusted, the differences in mortality diminished substantially. Consequently, age-adjusted rates permitted a more realistic comparison between the two states than crude rates.